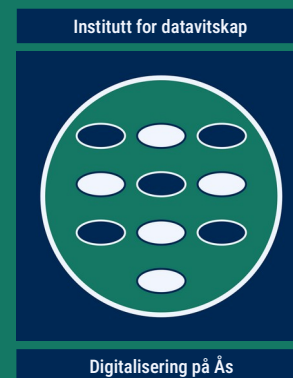
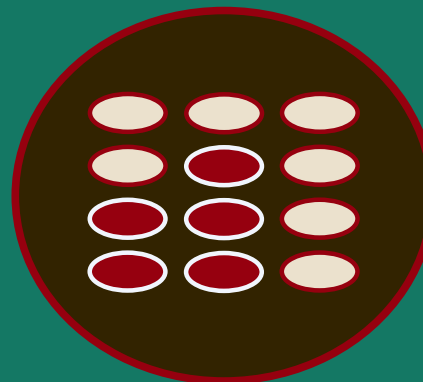


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INF230

Datahandtering og -analyse

1 Ontologiar

1.1 Den semantiske veven

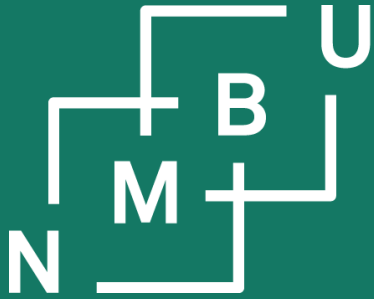
1.2 Vevontologispråket OWL

1.3 Terminologiutvikling

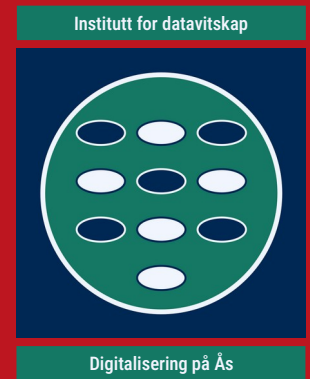
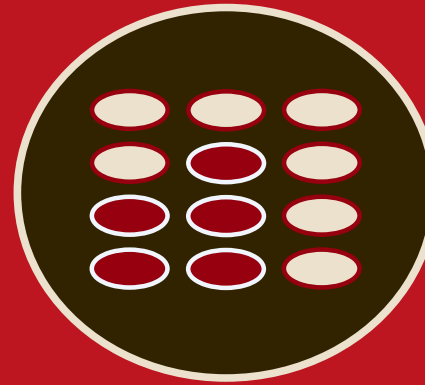
1.4 Ontologiutvikling

1.5 DOLCE, EMMO og BFO

1.6 Semantisk heterogenitet

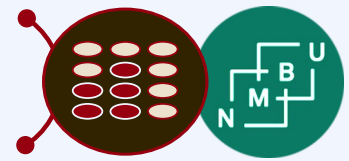


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1 Ontologiar

1.1 Metadata og semantisk vev



FAIR: From principles to technology¹

Findability

- F1. Globally unique **persistent identifiers (PID)**
- F2. **Enriched with metadata**
- F3. **Data identifier** included in **metadata**
- F4. **Registered in searchable platform**

Interoperability

- I1. **Formal language** used for **knowledge representation**
- I2. **Metadata** use **vocabularies** that are themselves FAIR
- I3. **Semantic web** principles, **data can refer to other data**

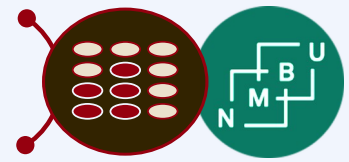
Accessibility

- A1. **Retrievable from PID** via a standard protocol
 - A1.1. Open and freely implementable protocol
 - A1.2. ... **authentication/authorization** if necessary
- A2. **Metadata** remain accessible (beyond data)

Reusability

- R1. **Metadata** include a plurality of accurate and relevant attributes
 - R1.1. Release data and **metadata** with an accessible **data usage license**
 - R1.2. **Data** are annotated with a detailed **provenance description**
 - R1.3. Relevant **disciplinary and community standards** are fulfilled

¹M. D. Wilkinson et al., "The FAIR Guiding Principles ...," doi:10.1038/sdata.2016.18, **2016**.

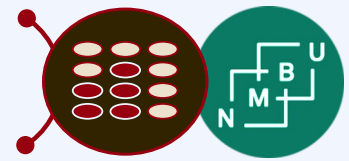


FAIR: From principles to technology¹

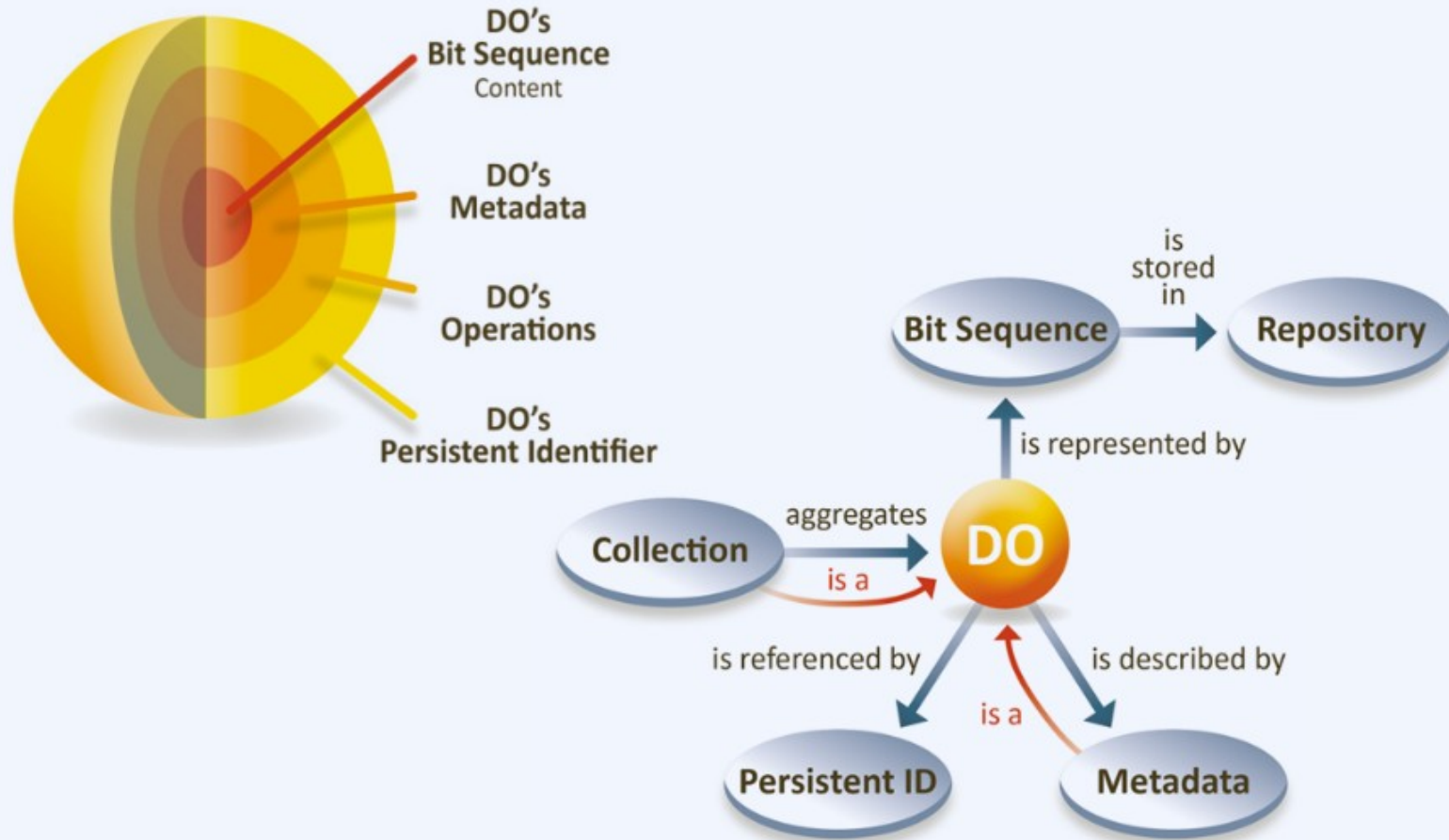
The principles precede implementation

These high-level FAIR guiding principles precede implementation choices and **do not suggest any specific technology**, standard, or implementation solution; moreover, the principles are not themselves a standard or a specification. They **act as a guide to data publishers and stewards to assist them in evaluating** whether their particular implementation choices are rendering their digital research artefacts findable, accessible, interoperable, and reusable. We anticipate that these high level principles will enable a broad range of integrative and exploratory behaviours, based on a wide range of technology choices and implementations.

¹M.D. Wilkinson, M. Dumontier, *et al.*, «The FAIR guiding principles for scientific data management and stewardship», *Sci. Data* **3**: 160018, doi:10.1038/sdata.2016.18, **2016**



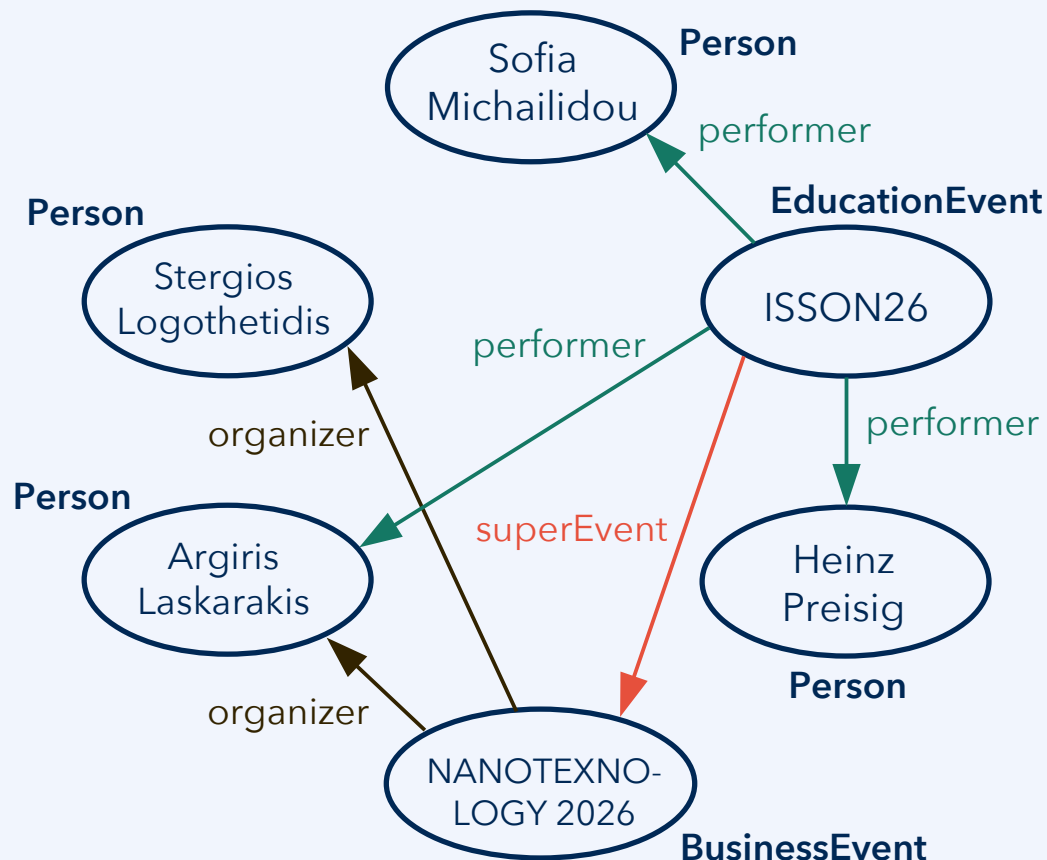
FAIR: From principles to technology¹



¹K. De Smedt, D. Koureas, P. Wittenburg, «FAIR digital objects for science: From data pieces to actionable knowledge units», *Publicat. (MDPI)* **8**(2): 21, doi:10.3390/publications8020021, **2020**

What is our course really about?

Modern knowledge bases represent knowledge about the state of affairs as **knowledge graphs**. These graphs are understood as part of one **semantic web**.



Knowledge graphs contain **individuals** (objects) as nodes.

They contain **relations** (binary predicates) as edges.

They may also visually represent the instantiation of **concepts** (classes).

What is our course really about?^{1, 2, 3}

1. Ontologies (2nd – 30th September 2026)

RDF schema, OWL ontologies, and how to develop and align OWL ontologies

2. Knowledge bases (1st – 14th October 2026)

Triple stores (graph DBs), SHACL, description logic, and reasoning

3. Structured Query Language (15th – 26th October 2026)

SQL as data definition, query, and manipulation language for relational databases

4. Relational database management systems (28th October – 9th November 2026)

Database design, relational model, relational algebra, relational calculus, data storage

5. Data systems analysis and architecture (9th – 26th November 2026)

Data management plans, knowledge based systems architectures and standards

INF131 (data management and logic) or equivalent is a prerequisite!

¹Course website (link)

²Course description (link)

³Course on Canvas (link)

Termliste

nn forhold *n.*

nn forholdstype *m.*

nn ikkje-unike namn *n.fl.*

nn kunnskapsgraf *m.*

nn openverdsprinsipp *n.*

nn persistent identifikator *m.* (PID)

nn ressurs *m.*

nn semantisk vev *m.*

nn trippel *m.*

en relationship

en relationship type / object property

en non-unique name assumption

en knowledge graph

en open world assumption

en persistent identifier (PID)

en resource

en semantic web

en triple

Semantic web data formats

JSON-LD

In order to make linked data in RDF more available to applications that use JSON, the W3C has recommended JSON-LD, JSON for Linked Data [Kellogg et al. 2014]. There is a direct correspondence between JSON-LD and RDF triples, making JSON-LD another serialization format for RDF.

N-Triples

The simplest form is called *N-Triples* and corresponds most directly to the raw RDF triples.

RDF/XML

While Turtle is convenient for human consumption and is more compact for the printed page, many Web infrastructures are accustomed to representing information in HTML or, more generally, XML. For this reason, the W3C historically started by recommending the use of an XML serialization of RDF called RDF/XML. The

Turtle/N3

In this book, we use a more compact serialization of RDF called *Turtle* which is itself a subset of a syntax called *N3*. Turtle combines the apparent display of triples from N-Triples with the terseness of CURIEs. We will introduce Turtle in this section

Allemang et al., avsnitt 3.8: Alternatives for serialization

Semantic web data formats

Basissyntaksen for triplar i TTL-notasjon er enkel: «subjekt predikat objekt .»

```
@prefix mfg: <http://www.WorkingOntologist.com/Examples/Chapter3/Manufacturing#>
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns>
```

It is quite common (especially after importing tabular data) to have several triples that share a common subject. Turtle provides for a compact representation of such data. It begins with the first triple in subject/predicate/object order, as before; but instead of terminating with a period, it uses a semicolon (;) to indicate that another triple with the same subject follows. For that triple, only the predicate and object need to be specified (since it is the same subject from before). The information in Tables 3.13 and 3.14 about Product1 and Product2 appears in Turtle as follows:

```
mfg:Product1 rdf:type mfg:Product;
              mfg:Product_Division "Manufacturing support";
              mfg:Product_ID "1";
              mfg:Product_Manufacture_Location "Sacramento";
              mfg:Product_ModelNo "ZX-3";
              mfg:Product_Product_Line "Paper Machine";
              mfg:Product_SKU "FB3524";
              mfg:Product_Available "23" .
```

When there are several triples that share both subject and predicate, Turtle provides a compact way to express this as well so that neither the subject nor the predicate needs to be repeated. Turtle uses a comma (,) to separate the objects. So the fact that Shakespeare had three children named Susanna, Judith, and Hamnet can be expressed as follows:

```
lit:Shakespeare b:hasChild b:Susanna , b:Judith , b:Hamnet .
```

There are actually three triples represented here—namely:

```
lit:Shakespeare b:hasChild b:Susanna .
lit:Shakespeare b:hasChild b:Judith .
lit:Shakespeare b:hasChild b:Hamnet .
```

Turtle provides some abbreviations to improve terseness and readability; in this book, we use just a few of these. One of the most widely used abbreviations is to use the word *a* to mean *rdf:type*. The motivation for this is that in common speech, we are likely to say, “Product1 is a Product” or “Shakespeare is a playwright” for the triples,

```
mfg:Product1 rdf:type mfg:Product .
lit:Shakespeare rdf:type lit:Playwright .
```

respectively. Thus we will usually write instead:

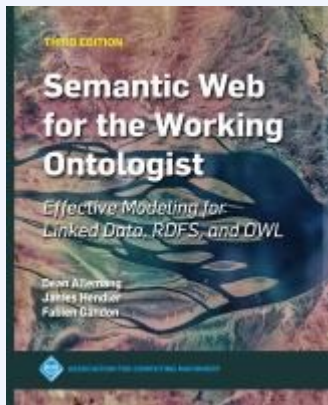
```
mfg:Product1 a mfg:Product .
lit:Shakespeare a lit:Playwright .
```

Allemang *et al.*, avsnitt 3.8: Alternatives for serialization

How we'll do the course

All required material will be made available on Canvas and on the course website. (Or there will be links to it on both.) You are fine just using Canvas.

There are three pensum books and two recommended books, officially.



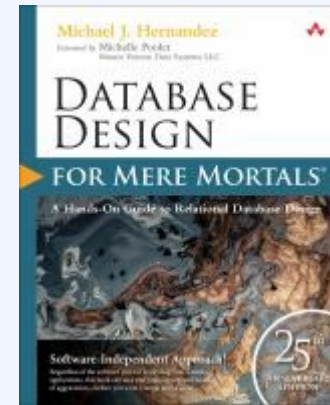
Allemang *et al.*



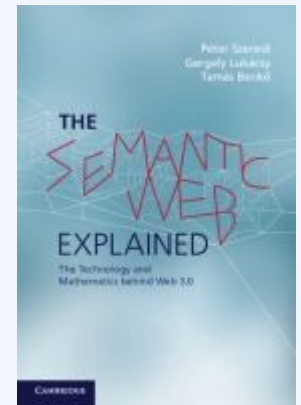
Keet



Silberschatz *et al.*



Hernandez



Szerefi *et al.*

How we'll do the course

All required material will be made available on Canvas and on the course website. (Or there will be links to it on both.) You are fine just using Canvas. *Of course, «private copies» from books will not be on my public website.*

There are three pensum books and two recommended books, officially. In reality, I will scan or copy the parts of the books that are pensum. They will be shared on Canvas under Filer / litteratur.

The exercise sessions will be done by Izza Qamar, and by you!
I try to get the lecture sessions recorded, but the IT needs to fix something ...

The language will be English as far as required, Norwegian as far as possible. All examination or oblig (*i.e.* exercise) material will be available in English. For such material, if it is made available in nynorsk, it will also be done in bokmål. For the exam, I still need to figure out if we do it in one or in three languages.

Exam and obligatory requirements

There will be character grades on the A – F scale, 100% from a written exam.

As an obligatory prerequisite for admissions to the exam, you need to solve five exercises. The exercises will be both on the website and on Canvas, but the submission will be on Canvas. You need to do all five (as a group).

The obligatory part is based on groups. There are up to 35 groups with up to four members each. You can form these groups yourselves. Remaining people will be assigned at random using the functionality for this on Canvas.

For the exercises, it is also obligatory that your group presents the solution to one of the problems that will be posed. We'll announce who presents what.

Unfortunately it is really impossible to work individually with now 132 people.



Exam and obligatory requirements

I have, as of now, no idea about the exact exam date.

The exercise submission deadlines and presentation sessions are as follows:

Worksheet	Deadline	Presentation
Oppgåve A	16th September, 8:00	16th September, 8:15 (øving A'')
Oppgåve B	30th September, 8:00	30th September, 8:15 (øving B'')
Oppgåve C	7th October, 8:00	7th October, 9:15 (øving C'')
Oppgåve D	5th November, 14:00	5th November, 15:15 (øving D'')
Oppgåve E	23rd November, 9:00	23rd November, 9:15 (øving A'')

We'll introduce the problems at dedicated sessions (labelled øving A', B', etc.).

Group formation

I have, as of now, no idea about the exact exam date.

The exercise submission deadlines and presentation sessions are as follows:

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Oppgåve D	5th November, 14:00	5th November, 15:15 (øving D'')
Oppgåve E	23rd November, 9:00	23rd November, 9:15 (øving E'')

We'll introduce the problems at dedicated sessions (labelled øving A', B', etc.).

Please think about forming a group right away, we'll get it done this week.

Semantic web principles

Semantic technology can facilitate the integration of data and software into a coherent framework, permitting multiple components to become interoperable.

concepts
(e.g. **Person**)

are instantiated by

individuals
(e.g. **Argiris Laskarakis**)

relations
(e.g. [has]
organizer)

are instantiated by

relationships
(e.g. NANOTECHNOLOGY
[has] organizer Argiris L.)

RDF is the Resource Description Framework, which specifies the semantic web. In this context, a **resource** is any of the following:

an **individual** (i.e., object); a **concept** (i.e., class); a property/**relation**.

Resources are referenced by using **Internationalized Resource Identifiers (IRIs)**.

Semantic web principles

Triples: Individual Relation Individual. (Subject Predicate Object.)

(1) Frank is father of Robert.

Human is a concept.

(2) Frank, Robert, Nick each are a Human.

Q: "Is Nick the father of Robert?"

Cardinality restriction:

A: "No idea, we cannot tell."

Every Human has exactly 1 father.

Principle: Non-unique name assumption

It is normal for one real-world entity (or generally, one "resource") to have multiple identifiers (here, IRIs, internationalized resource identifiers). For example, they can be referred to in different namespaces. So "namespace_a:CH4" and "namespace_b:methane" could be the same object.

Similarly, "Nick" and "Frank" could in principle be the same object.

Semantic web principles

Triples: Individual Relation Individual. (Subject Predicate Object.)

(1) Frank is father of Robert.

(2) Frank is different from Nick.

Q: "Is Nick the father of Robert?"

A: "No, he is not."

(3) Frank is father of Anna.

Q: "How many children does Frank have?"

A: "At least two."

Human is a concept.

Frank, Robert, etc., are Humans.

Cardinality restriction:

Every Human has exactly 1 father.

Anna is different from Robert.

"How many different X are there such that Frank is_father_of X?"

Principle: Open world assumption

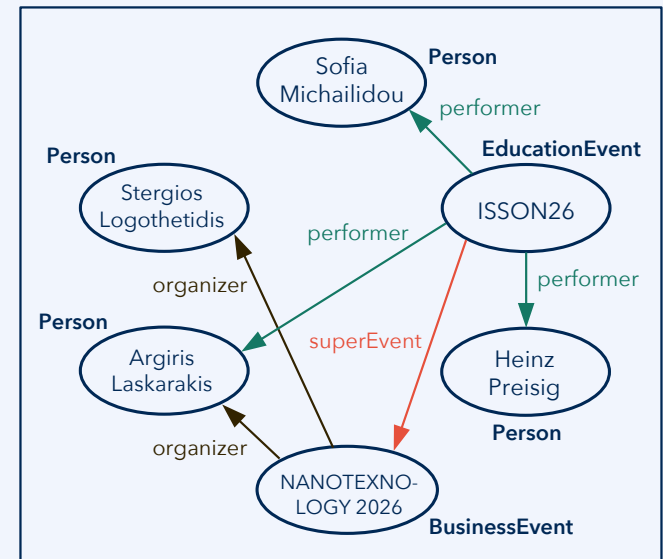
Since relevant information may distributed over the semantic web, rather than from the presently considered source only, **available knowledge is assumed to be incomplete**. (Contrast this with a closed, monolithic database architecture.)

Serialization of knowledge graphs

JSON-LD: Extension of JSON to deal with linked data (knowledge graphs)

JSON: JavaScript Object Notation

```
{
  "@context": {
    "isson": "http://batcat.info/isson/data#",
    "schema": "https://schema.org/"
  },
  "@id": "isson:ISSON26",
  "@type": "schema:EducationEvent",
  "schema:performer": [
    {
      "@id": "isson:Argiris_Laskarakis",
      "@type": "schema:Person"
    }, {
      "@id": "isson:Sofia_Michailidou",
      "@type": "schema:Person"
    }, ...
  ],
  "schema:superEvent": {
    "@id": "isson:NANOTEXNOLOGY_2026",
    "@type": "schema:BusinessEvent",
    "schema:organizer": [
      {
        "@id": "isson:Argiris_Laskarakis"
      }, {
        "@id": "isson:Stergios_Logothetidis",
        "@type": "schema:Person"
      }
    ]
  }
}
```



Serialization of knowledge graphs

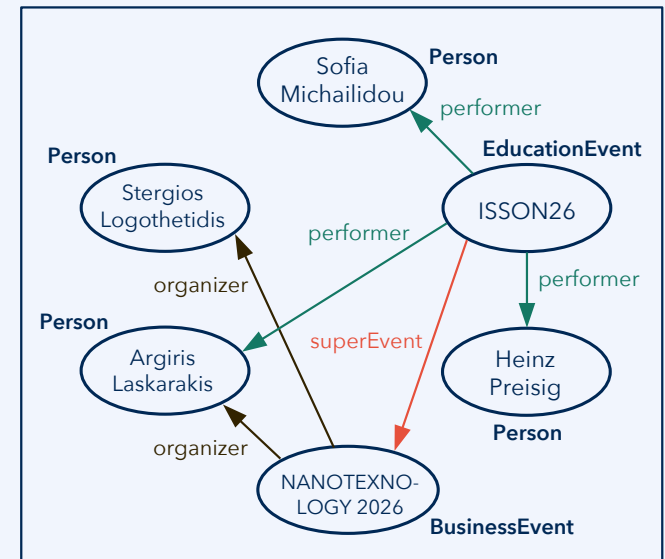
Terse triple language (TTL), also called "Turtle," for our example scenario:

```
@base <http://batcat.info/isson/data#>.  
@prefix schema: <https://schema.org/>.
```

```
<Argiris_Laskarakis> a schema:Person.  
<Stergios_Logothetidis> a schema:Person.  
<Sofia_Michailidou> a schema:Person.  
<Heinz_Preisig> a schema:Person.
```

```
<ISSON26> a schema:EducationEvent;  
  schema:performer <Argiris_Laskarakis>, <Sofia_Michailidou>,  
    <Heinz_Preisig>;  
  schema:superEvent <NANOTEXNOLOGY_2026>.
```

```
<NANOTEXNOLOGY_2026> a schema:BusinessEvent;  
  schema:organizer <Argiris_Laskarakis>, <Stergios_Logothetidis>.
```



SPARQL data query language

6.3 SPARQL—Query Language for RDF

The syntax of SPARQL actually looks like Turtle. So these examples really look more like this:

Tell: `:JamesDean :playedIn :Giant .`

Ask: `:JamesDean :playedIn ?what .`

Answer: `:Giant`

Ask: `?who :playedIn :Giant .`

Answer: `:JamesDean`

Ask: `:JamesDean ?what :Giant .`

Answer: `:playedIn`

Before we go further, let's talk a bit about the syntax of a SPARQL query. We'll start with a simple form, the SELECT query. Readers familiar with SQL will notice a lot of overlap with SPARQL syntax (for example, keywords like SELECT and WHERE). This is not coincidental; SPARQL was designed to be easily learned by SQL users.

RDF (and SPARQL) deals well with multiple values. If we TELL the system that James Dean played in multiple movies, we can do this without having to make any considerations in the representation:

Tell: `:JamesDean :playedIn :Giant .
:JamesDean :playedIn :EastOfEden .
:JamesDean :playedIn :RebelWithoutaCause .`

Now if we ASK a question with SPARQL

Query 6.1 James Dean Played In

Ask: `SELECT ?what WHERE { :JamesDean :playedIn ?what }`

Answer: `:Giant , :EastOfEden , :RebelWithoutaCause .`

The WHERE clause of a SPARQL query can be seen as a graph pattern, that is, a pattern that is matched against the data graph.

SPARQL data query language

Query structure vs. data structure

In Figure 6.4, we saw how the graph pattern in a query looks a lot like the data graph it matches against. This is true of graph patterns in general. The complexity of a question that can be specified with SPARQL is limited only by the complexity of the data. We could, for instance, ask about actresses who played in a movie with James Dean who themselves were in a movie directed by John Ford:

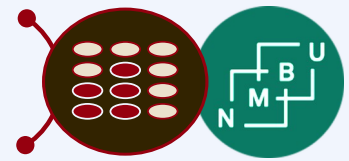
```
SELECT ?actress ?movie
WHERE { :JamesDean :playedIn ?movie .
        ?actress :playedIn ?movie .
        ?actress a :Woman .
        ?actress :playedIn ?anotherMovie .
        ?anotherMovie :directedBy :JohnFord . }
```

Answer:

```
?actress
NatalieWood
CarrollBaker
```

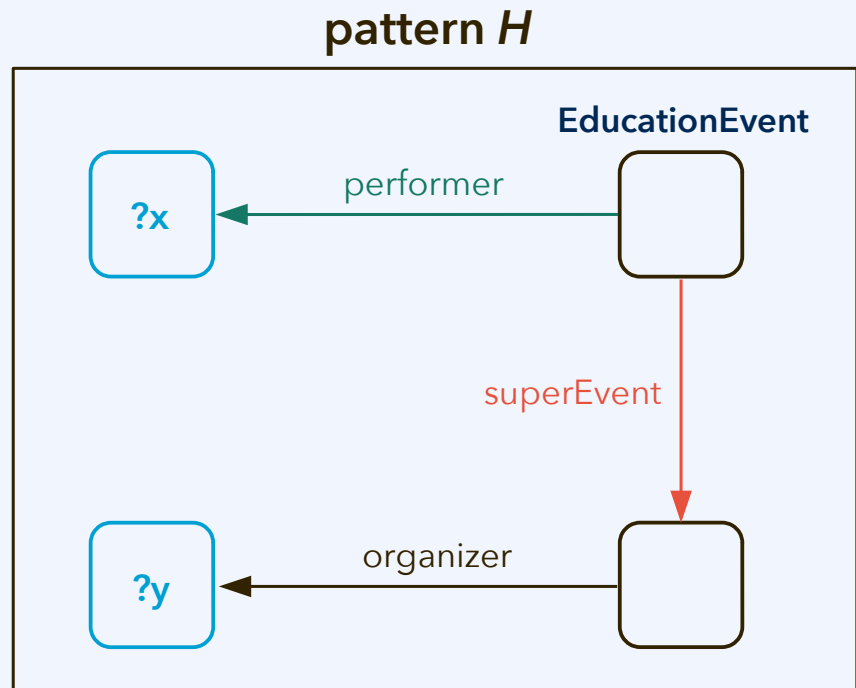
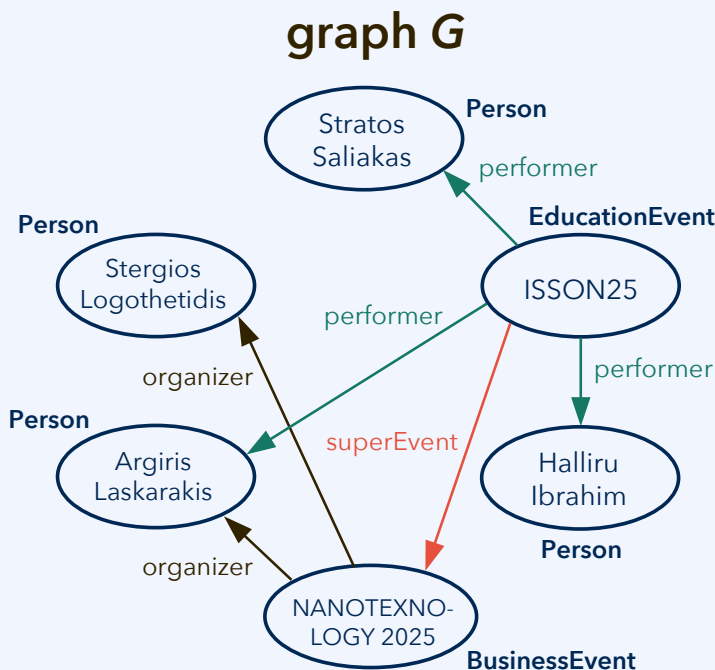
In the text version of the query, we often see the same question word appear in multiple triples, and some of them even refer to the same kind of thing (?movie, ?anotherMovie). In the graph version, we see that these are the points where two triples must refer to the same thing.

Allemang et al., avsnitt 6.3: SPARQL: Query language for RDF

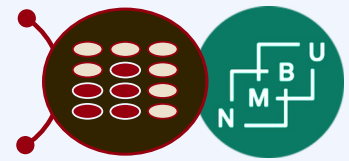


Querying a SPARQL end point

Given a graph G and a pattern H , does G contain a subgraph isomorphic to H ?



The **semantics of SPARQL** is given by the correct response to a SPARQL query, which consists of a *table with all matching valuations of the selected wildcards*.



Querying a SPARQL end point

SPARQL is a recursive acronym: "SPARQL Protocol and RDF Query Language."
An interface that can handle SPARQL queries is called a *SPARQL end point*.

The **syntax of SPARQL** is given by

- **RDF triples** in **TTL notation**;
- Some elements of the triples are **wildcards**, i.e., **free variables**.

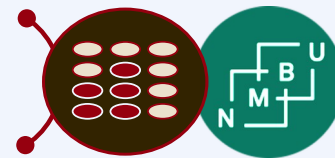
prefix schema: <https://schema.org/>

```
SELECT ?x ?y
WHERE {
  ?school schema:performer ?x.
  ?school rdf:type schema:EducationEvent.
  ?school schema:superEvent ?conference.
  ?conference schema:organizer ?y.
}
```

"Produce a table with summer school lecturers in the first column and all the corresponding responsible conference organizers in the second column."

The **semantics of SPARQL** is given by the correct response to a SPARQL query, which consists of a *table with all matching valuations of the selected wildcards*.

Glossar



kunnskapsgraf

Definisjon: graf der nodar er individ (objekt) og kantar er binære forhold mellom individ

semantisk vev

Definisjon: «del av internett med koplede datasett som [...] består av enkle påstander som er forståelige for maskiner» (A. Gjersdal i SNL)

ressurs

Forklaring: Den semantiske veven består av tre slags ressursar: **Omgrep** (også "konsept" eller "klasser"), **forholdstypar** (også "relasjonar" eller "roller") og **individ** (også "objekt"). Kvar ressurs blir identifisert gjennom ein **internasjonalisert ressursidentifikator (IRI)**.

ikkje-unike namn

Definisjon: grunnregel på semantisk vev som inneber at det er mogleg å referere til den same ressursen gjennom fleire forskjellige internasjonaliserte ressursidentifikatorar

Glossar

trippel

Forklaring: «Metadata i den semantiske veven uttrykkes i korte setninger som kalles trippler. Dette er enkle påstander som lages ut fra den åpne standarden Resource Description Framework (RDF), og består av tre elementer: **subjekt**, **predikat** og **objekt**» (SNL).

forhold – in English: **relationship**

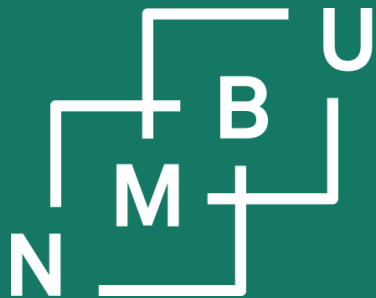
Definisjon: elementært fakt om konkrete objekt

forholdstype – in English: **object property / relationship type / relation**

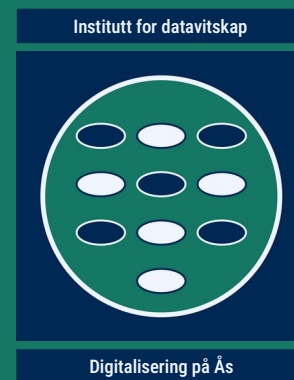
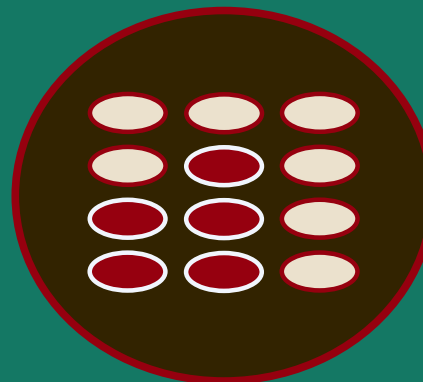
Definisjon: tilknytning av klasser som beskriv korleis individ som høyrer til dei respektive klassene, kan forhalde seg til einannan; tilsvarar som regel eit verb i menneskeleg språk

openverdsprinsipp

Definisjon: grunnregel anvend på kunnskapsbasar og semantisk vev, som tilseier at det ved evaluering av spørjingar, logisk deduksjon, mm., er underforstått at den umiddelbart tilgjengelege informasjonen kan vera ufullstendig



Noregs miljø- og
biovitenskaplege
universitet



INF230

Datahandtering og -analyse

1 Ontologiar

1.1 Den semantiske veven

1.2 Vevontologispråket OWL

1.3 Terminologiutvikling

1.4 Ontologiutvikling

1.5 DOLCE, EMMO og BFO

1.6 Semantisk heterogenitet